SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

C0 3 ASSESSMENT TOOL 2

Course Code:	ITA0609	**Course Title:**	Machine Learning
CO Assessed:	CO 3	**Assessment:**	Assessment Tool 2
Weightage:	40% of CIA	**Total Marks:**	
CO1 Statement:	*To acquire a foundational understanding of key machine learning concepts including version spaces, candidate elimination, inductive bias, and heuristic search (BL1)*		
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Section / Batch:	SLOT A	**Date:**	07.08.26

Code of Conduct

I (N.kishan kumar-192424404) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Instructions.

- This test contains 5 Short questions, each carrying 10 mark.
- All questions are application-based and require analytical thinking and practical implementation. · No negative marking. Unanswered questions carry zero marks.
- Provide suitable algorithms, logic, calculations, or case-based solutions wherever required.

CASE STUDY 1: Smart Healthcare System Using Bayesian Belief Networks

1. Introduction

A smart healthcare monitoring system uses wearable sensors to continuously collect physiological parameters such as heart rate, blood pressure, body temperature, oxygen saturation (SpO_2), and respiratory rate. Since sensor readings may contain noise and patient conditions are uncertain, Bayesian Belief Networks (BBNs) and probability learning can be used to detect abnormal health conditions.

A Bayesian Belief Network represents relationships among physiological parameters using a directed acyclic graph (DAG). Each node represents a variable, while the connections represent conditional dependencies.

2. Parameters Used

Parameter	Normal Range	Possible Abnormality
Heart Rate	60–100 bpm	Bradycardia / Tachycardia
SpO_2	95–100%	Hypoxemia
Body Temperature	36.1–37.5°C	Fever / Hypothermia
Blood Pressure	Around 120/80 mmHg	Hypotension / Hypertension
Respiratory Rate	12–20 breaths/min	Respiratory abnormality

3. Bayesian Belief Network Model

A possible dependency structure is:

Sensor Noise → Observed Sensor Values

The joint probability can be represented as:

For example:

where **D** represents the presence of an abnormal health condition.

4. Conditional Dependencies

Physiological parameters are not completely independent. For example, a respiratory problem may cause SpO_2 to decrease and respiratory rate to increase. Similarly, fever may increase both body temperature and heart rate.

BBNs explicitly represent these dependencies instead of treating each sensor reading separately. Therefore, the system can distinguish between a single unusual measurement and a combination of measurements indicating a genuine medical anomaly.

Example

Suppose the sensor reports:

Vital Sign	Reading	Status
Heart Rate	118 bpm	High

SpO_2	89%	Low
Temperature	38.5°C	High
Respiratory Rate	25/min	High

The combined evidence gives a much higher probability of an abnormal condition than any single parameter alone.

5. Handling Sensor Noise and Uncertainty

Wearable sensors may produce inaccurate readings because of movement, poor skin contact, hardware limitations, transmission errors, or environmental conditions.
Bayesian probability handles this uncertainty by assigning probabilities instead of making rigid decisions.

For example:
Instead of immediately declaring an emergency, the system considers the reliability of the sensor and evidence from other parameters.
Sensor reliability can also be represented as a separate node in the Bayesian network. Repeated readings and historical patient data can further reduce the effect of temporary sensor noise.

6. Probability Learning

Historical patient records can be used to estimate the Conditional Probability Tables (CPTs).
For example:

SpO_2	Heart Rate	P(Abnormal Condition)
Normal	Normal	0.05
Low	Normal	0.45
Normal	High	0.35
Low	High	0.90

As additional patient data becomes available, these probabilities can be updated, allowing the model to improve over time.

7. Performance Evaluation

The system can be evaluated using:
Accuracy
Precision
Recall/Sensitivity
F1 Score

In healthcare monitoring, recall is particularly important, because a false negative may result in a dangerous medical condition being missed. However, excessive false positives can create alarm fatigue.

8. Challenges in Real-Time Deployment

Major challenges include sensor noise, missing readings, limited wearable battery life, network delays, patient-to-patient physiological differences, large amounts of streaming data, privacy and security requirements, and the need for very fast inference.

Another important problem is alarm fatigue. If the system generates too many false warnings, healthcare professionals may start ignoring alerts. Therefore, the probability threshold must balance sensitivity and false-alarm rate.

9. Conclusion

Bayesian Belief Networks provide an effective framework for smart healthcare monitoring because they represent conditional dependencies, uncertainty, sensor reliability, and probabilistic relationships among vital signs. By combining evidence from multiple sensors, the system can provide more reliable anomaly detection than fixed threshold-based methods.

CASE STUDY 2: Cybersecurity Intrusion Detection Using Probabilistic Learning

1. Introduction

Modern computer networks face attacks such as DoS/DDoS attacks, port scanning, brute-force login attempts, malware communication, and unauthorized access. A cybersecurity intrusion detection system can use Naïve Bayes, Expectation-Maximization (EM), and Bayesian learning to classify network activity as normal or abnormal.
Probabilistic learning is useful because network behavior is uncertain and constantly changing.

2. Network Features

The system can analyze features such as:

Feature	Purpose
Source/Destination IP	Identify communication endpoints
Protocol	TCP, UDP, ICMP etc.
Packet Size	Identify abnormal traffic patterns
Connection Duration	Detect unusual sessions
Failed Login Count	Detect brute-force attacks
Packet Frequency	Detect flooding attacks
Port Number	Identify suspicious services
Traffic Volume	Detect unusual network usage

3. Naïve Bayes Classifier

Naïve Bayes calculates the probability that network traffic belongs to a particular class.
Using Bayes' theorem:
where:

- = class such as Normal or Intrusion
- = observed network features
- = prior probability
- = likelihood
- = posterior probability.

Naïve Bayes assumes that features are conditionally independent given the class:
The traffic is assigned to the class having the highest posterior probability.

Example

Suppose a connection has:

- Extremely high packet frequency
- Many failed login attempts
- Unusual destination port
- Short repeated connections

The model may calculate:
Therefore, the connection is classified as an intrusion.

4. EM Algorithm for Hidden Patterns

Cybersecurity datasets may contain missing information or unlabeled traffic. The Expectation-Maximization (EM) algorithm can discover hidden patterns from such incomplete data.

EM consists of two major steps:

E-Step – Expectation

Estimate probabilities for hidden or missing variables using the current model parameters.
Here, represents hidden variables.

M-Step – Maximization

Update the model parameters to maximize the expected likelihood.
These two steps are repeated until the model converges.

For example, if some traffic records do not have labels indicating whether they are attacks, EM can estimate their likely classes and improve the model parameters using both labeled and unlabeled observations.

5. Bayesian Learning

Bayesian learning continuously updates the probability of network attacks as new evidence becomes available.
where represents an attack hypothesis and represents observed network data.
The posterior probability obtained from previous observations can become the prior probability for future observations. Therefore, the intrusion detection model can adapt to changing network conditions.

6. Handling Hidden Patterns and Incomplete Data

Problem	Probabilistic Solution
Missing network values	EM estimates missing information
Unlabeled traffic	EM identifies probable hidden classes
Uncertain attack behavior	Bayesian probabilities represent uncertainty
New traffic patterns	Bayesian learning updates beliefs
Large feature sets	Naïve Bayes provides fast classification
Noisy observations	Probability thresholds reduce rigid decisions

Combining these methods creates a more flexible intrusion detection system.

7. Reducing False Alarms

Traditional rule-based intrusion detection systems may classify any unusual traffic as malicious, producing many false alarms.

Probabilistic systems instead calculate:

An alert can be generated only when:

For example, if the threshold is **0.85**, traffic with an attack probability of 0.60 may be monitored rather than immediately generating a high-priority alert.

Thresholds can be adjusted according to the security requirements of the organization.

8. Performance Evaluation

Important measures are:

A successful system should have high detection rate and precision while maintaining a low false-positive rate.

9. Dynamic Environment Challenges

Network traffic changes continuously because of new applications, users, devices and attack techniques. This produces **concept drift**, where previously learned patterns may no longer accurately represent current traffic.

Other challenges include encrypted traffic, zero-day attacks, highly imbalanced datasets, large network traffic volumes, adversarial behavior and the computational cost of continuous learning.

The model should therefore be periodically updated with recent traffic patterns.

10. Conclusion

A probabilistic intrusion detection system combining Naïve Bayes, EM and Bayesian learning can effectively identify suspicious network activity. Naïve Bayes provides fast classification, EM handles hidden and incomplete information, and Bayesian learning allows probabilities to adapt as new network evidence appears. This combination can improve detection accuracy while reducing unnecessary false alarms.

CASE STUDY 3: Crop Yield Prediction Using Bayesian Inference

1. Introduction

Crop yield depends on uncertain environmental and agricultural factors including **rainfall,** temperature, soil fertility, humidity, irrigation, crop variety, fertilizer application and pest occurrence.

Since these variables contain uncertainty, Bayesian inference and probability learning can be used to estimate the probability of different crop-yield levels.
The system can help farmers make better decisions regarding planting, irrigation, fertilizer usage and harvesting.

2. Important Input Parameters

Parameter	**Influence on Crop Yield**
Rainfall	Determines water availability
Temperature	Influences crop growth
Soil Quality	Determines nutrient availability
Soil Moisture	Indicates water availability
Fertilizer	Supports plant growth
Humidity	Influences crop and disease conditions
Pest/Disease	Can decrease yield
Crop Variety	Determines productivity
Seasonal Conditions	Influence overall growing conditions

The target variable may be classified as Low Yield, Medium Yield or High Yield, or represented as a continuous value such as tonnes per hectare.

3. Bayesian Predictive Model

Bayes' theorem is:
where:

- = crop yield
- = environmental evidence
- = prior probability of yield
- = likelihood of environmental conditions
- = posterior probability of yield.

For example:
provides the probability of achieving high crop yield given current environmental conditions.

4. Model Structure

A simple Bayesian network can be represented as:
Season → Rainfall
Season → Temperature
Soil Quality → Nutrient Availability

Rainfall → Soil Moisture
Irrigation → Soil Moisture
Temperature → Crop Growth
Soil Moisture → Crop Growth
Nutrient Availability → Crop Growth
Pest/Disease → Crop Growth
Crop Growth → Yield
This network captures both direct and indirect relationships affecting agricultural production.

5. Influence of Prior Agricultural Knowledge
A major advantage of Bayesian inference is its ability to incorporate **prior knowledge**.

Suppose historical records indicate:
After observing favorable rainfall, temperature and soil conditions:
Therefore, the model combines historical knowledge with current observations.
Prior knowledge may come from historical crop records, agricultural experts, soil studies, regional climate information and previous seasonal performance.

As new observations become available:
The posterior distribution can then be used as prior knowledge for future predictions.

6. Effect of Seasonal Variation
Seasonal variation has a major influence on crop production. Rainfall and temperature patterns differ between **monsoon, summer and winter seasons**, and different crops respond differently to these conditions.

Instead of assuming a fixed rainfall-yield relationship, the Bayesian model can calculate:

For example, 100 mm of rainfall may have different effects depending on crop growth stage and season.
Season-specific historical data therefore improves prediction reliability.

7. Example Prediction
Consider the following conditions:

Factor	Observation
Rainfall	Adequate
Soil Quality	High
Temperature	Optimal
Soil Moisture	Adequate
Pest Risk	Low

After Bayesian inference, assume:
The farmer can therefore expect a high yield with **78% probability**, while still understanding that uncertainty exists.

8. Model Reliability
For categorical yield prediction, performance can be measured using accuracy, precision, recall and F1-score.
For numerical crop-yield prediction, useful metrics include:
Mean Absolute Error
Root Mean Square Error
Lower MAE and RMSE values indicate better prediction accuracy.
Reliability should also be tested across different seasons, geographical regions, soil types and crop varieties rather than using only one dataset.

9. Impact on Farmer Decision-Making
Crop-yield predictions can help farmers choose suitable crops, determine planting time, optimize irrigation, plan fertilizer usage, prepare for low-yield conditions and estimate expected production.

For example, if the model predicts a high probability of low yield because of insufficient rainfall, the farmer may increase irrigation or select a drought-resistant crop.

10. Impact on Policy Planning
At a larger scale, predicted crop yields can help governments and agricultural organizations with food supply planning, crop insurance, drought management, storage planning, import/export decisions, subsidy allocation and agricultural resource distribution.
Probabilistic predictions are especially useful because policymakers can plan according to multiple possible outcomes rather than relying on a single fixed estimate.

11. Challenges
The main challenges include incomplete weather data, unpredictable climate change, differences in soil properties, pest outbreaks, inaccurate sensor measurements, regional variation, limited historical data and sudden extreme weather events.
The model must therefore be regularly updated with new agricultural and climate information.

12. Conclusion
Bayesian inference provides a reliable approach to crop-yield prediction under uncertain environmental conditions. It combines prior agricultural knowledge, current environmental observations and seasonal information to produce probabilistic yield estimates. The resulting predictions can support better decisions for both individual farmers and government policymakers.

Overall Comparison of the Three Case Studies

Case Study	Main Technique	Major Uncertainty	Main Benefit
Smart Healthcare	Bayesian Belief Network	Sensor noise and physiological variation	Early anomaly detection

Cybersecurity	Naïve Bayes + EM + Bayesian Learning	Hidden/incomplete network data	Intrusion detection with fewer false alarms
Agriculture	Bayesian Inference	Weather and seasonal uncertainty	Reliable crop-yield prediction

Overall Learning Outcome

All three case studies demonstrate that probability learning is highly suitable for real-world systems where information is uncertain or **incomplete**. Bayesian methods represent uncertainty mathematically and update predictions when new evidence becomes available. In healthcare, this enables reliable patient monitoring; in cybersecurity, it supports adaptive intrusion detection; and in agriculture, it improves crop-yield forecasting and decision-making.